are mixed in the mixing chamber of the welding torch. Flame is produced at the tip of the torch when the gases are ignited. There are two control valves present in the torch to control the quantity of oxygen and acetylene. By this control, the grade of the flame can be altered. The size of the flame is altered to suit the thickness of the metal parts.

### Goggles

Oxy acetylene flame emits ultraviolet and infrared rays. These rays are highly harmful to bare eyes. In order to protect the eyes of the welder, goggles should be used by him.



Goggles

### Welding gloves

Protective hand gloves are used by the operator to prevent possible damages that may be caused by high temperatures and metal splashes during welding.



Gloves

### Spark lighter

Spark lighter is used to ignite the oxy-acetylene gas at the tip of the welding torch.



Spark Lighter

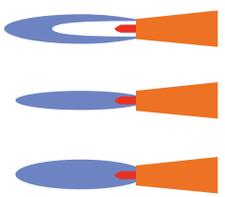
### Wire brush

Wire brushes are useful for cleaning the weld before and after the welding process.

## 7.12 Types of gas flame

The size of the flame can be altered by varying the ratio of oxygen and acetylene. By doing so, the following three types of flames are obtained.

1. Neutral flame
2. Carburising flame
3. Oxidising flame



### Neutral flame

The supply of equal quantities of oxygen and acetylene produces neutral flame. There are two zones in this flame-1. Sharp and bright inner cone and 2. Bluish outer cone. The temperature of the inner cone will be around $3200^0$C. This neutral flame is generally used as it will not cause any chemical reaction upon the heated metal.



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